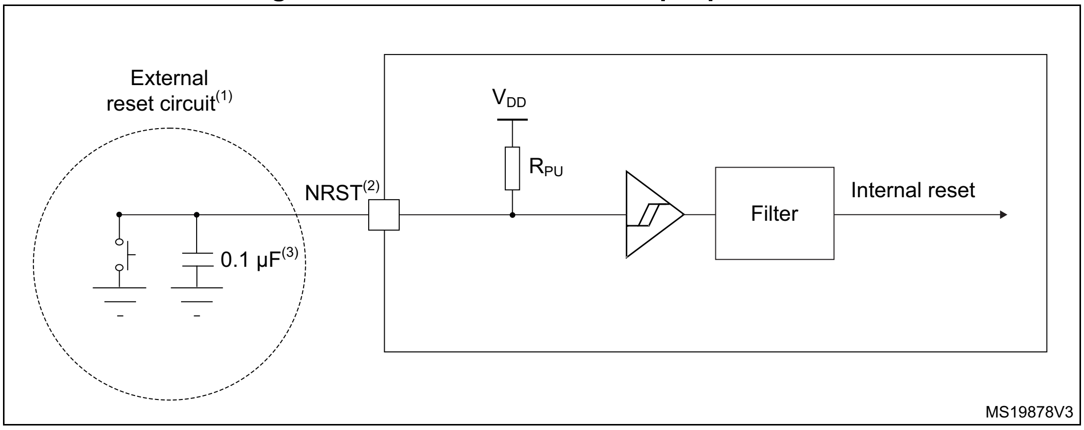
Figure 21. Recommended NRST pin protection



1. The reset network protects the device against parasitic resets.
2. The user must ensure that, upon power-on, the level on the NRST pin can exceed the minimum $V_{IH(NRST)}$ level specified in *Table 53: NRST pin characteristics*. Otherwise, the device does not exit the power-on reset. This applies to any NRST configuration set through the NRST_MODE[1:0] bitfield, the GPIO mode inclusive.
3. The external capacitor on NRST must be placed as close as possible to the device.

5.3.15 Analog-to-digital converter characteristics

Unless otherwise specified, the parameters given in *Table 54* are preliminary values derived from tests performed under ambient temperature, f_{PCLK} frequency and V_{DDA} supply voltage conditions summarized in *Table 24: General operating conditions*.

Note: It is recommended to perform a calibration after each power-up.

Table 54. ADC characteristics(1)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Analog supply voltage	-	2.0	-	3.6	V
V_{REF+}	Positive reference voltage	-	2	-	V_{DD}	V
f_{ADC}	ADC clock frequency	-	0.14	-	35	MHz
f_s	Sampling rate	12 bits	-	-	2.50	MSps
		10 bits	-	-	2.92	
		8 bits	-	-	3.50	
		6 bits	-	-	4.38	
f_{TRIG}	External trigger frequency	f_{ADC} = 35 MHz; 12 bits	-	-	2.33	MHz
		12 bits	-	-	$f_{ADC}/15$	
V_{AIN}	Conversion voltage range	-	0	-	V_{REF+}(2)	V
R_{AIN}	External input impedance	-	-	-	50	kΩ